

Name:SyedSalman.S

Reg no:192524076

Course code:DSA0604

Course name: Data Handling and Visualization

Set 1: Monthly Sales Data

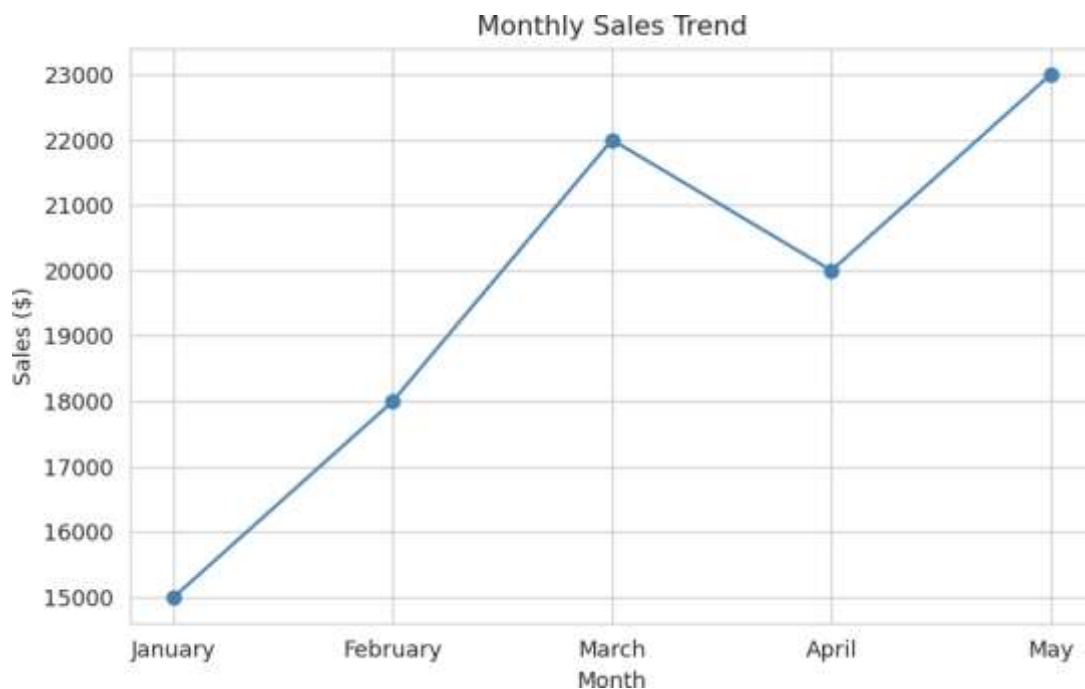
Question 1

Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

R Code

```
# 1. Line chart - monthly sales
ggplot(df, aes(x = Month, y = Sales, group = 1)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_point(color = "darkblue") +
  labs(title = "Monthly Sales Trend", x = "Month", y = "Sales ($)") +
  theme_minimal()
ggsave("line_chart_monthly_sales.png", width = 7, height = 5)
```

Output



Question 2

Generate a bar chart to display the top-selling products for the year. Label the chart elements.

R Code

```
# 2. Bar chart - top-selling products for the year
ggplot(df, aes(x = Product, y = ProductSales, fill = Product)) +
  geom_bar(stat = "identity") +
  labs(title = "Top-Selling Products for the Year", x = "Product", y = "Sales ($)") +
  theme_minimal()
ggsave("bar_chart_top_products.png", width = 7, height = 5)
```

Output



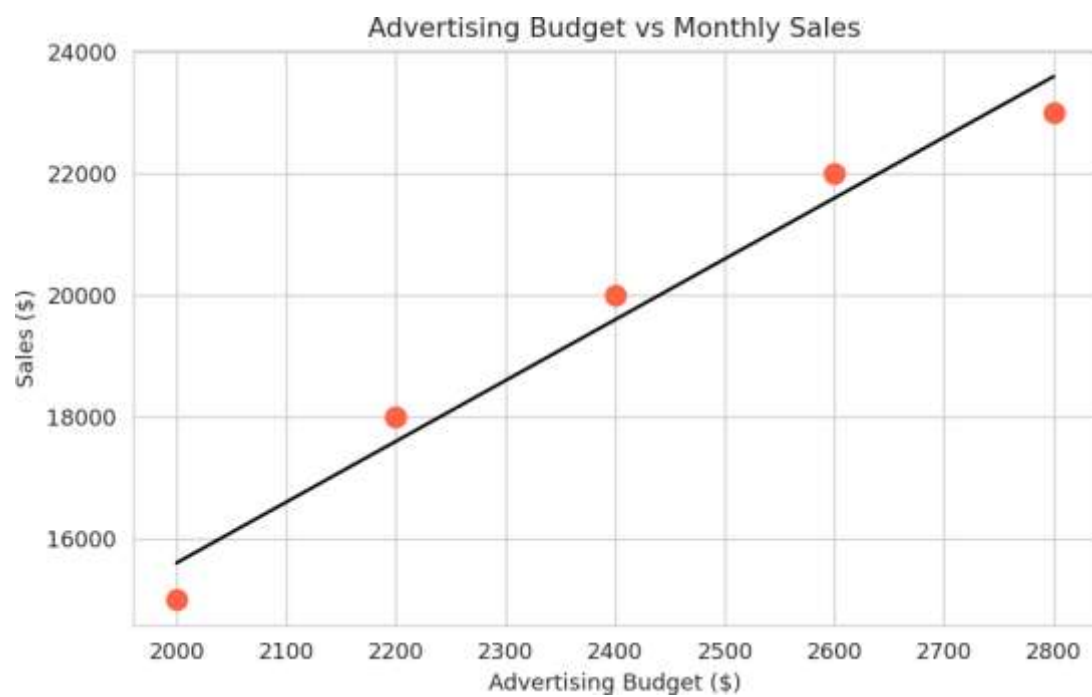
Question 3

Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.

R Code

```
# 3. Scatter plot - advertising budget vs monthly sales
ggplot(df, aes(x = AdvertisingBudget, y = Sales)) +
  geom_point(color = "tomato", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(title = "Advertising Budget vs Monthly Sales", x = "Advertising Budget ($)", y =
"Sales ($)") +
  theme_minimal()
ggsave("scatter_ad_budget_vs_sales.png", width = 7, height = 5)
cat("Insight: There is a positive relationship - as advertising budget increases,
monthly sales tend to increase.\n")
```

Output



Set 2: Customer Feedback Analysis

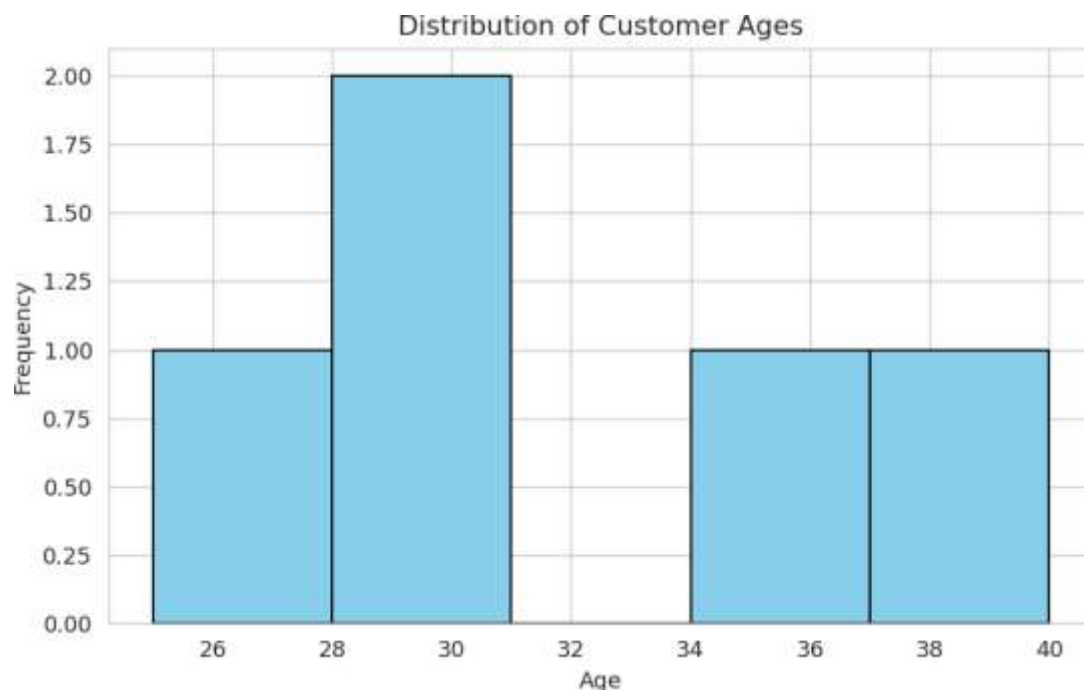
Question 1

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

R Code

```
# 1. Histogram - distribution of customer ages
ggplot(df, aes(x = Age)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Customer Ages", x = "Age", y = "Frequency") +
  theme_minimal()
ggsave("histogram_customer_ages.png", width = 7, height = 5)
```

Output



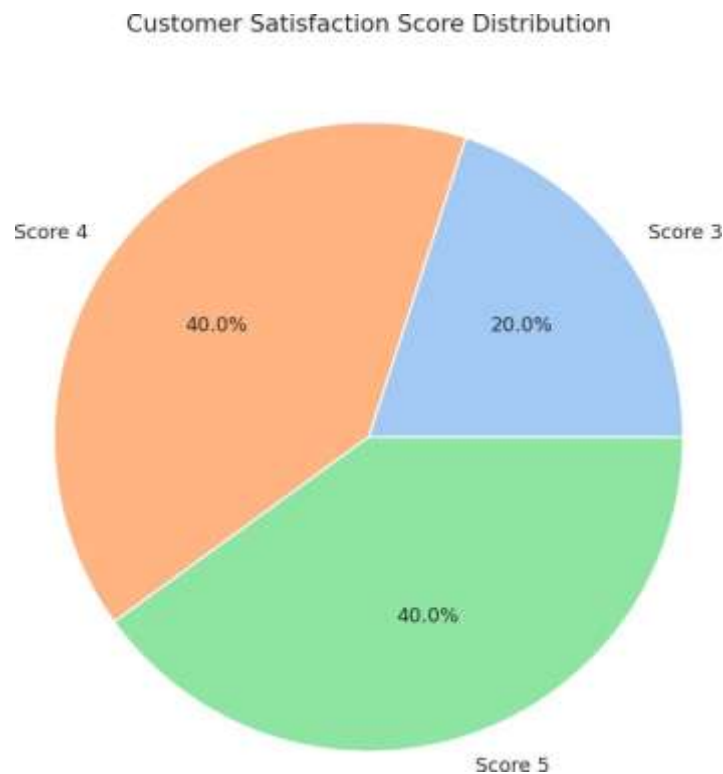
Question 2

Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

R Code

```
# 2. Pie chart - distribution of satisfaction scores
score_counts <- df %>% count(SatisfactionScore)
ggplot(score_counts, aes(x = "", y = n, fill = factor(SatisfactionScore))) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(title = "Customer Satisfaction Score Distribution", fill = "Score") +
  theme_void()
ggsave("pie_chart_satisfaction.png", width = 7, height = 5)
```

Output



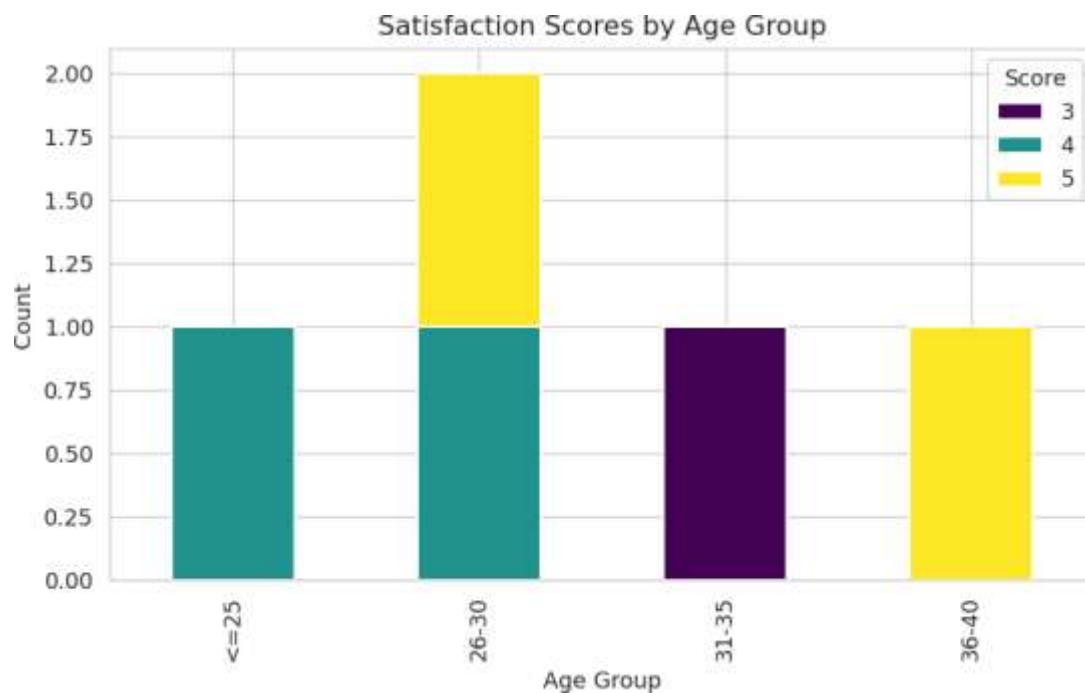
Question 3

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

R Code

```
# 3. Stacked bar chart - satisfaction scores by age group
df$AgeGroup <- cut(df$Age, breaks = c(0, 25, 30, 35, 40, 100),
  labels = c("<=25", "26-30", "31-35", "36-40", "40+"))
ggplot(df, aes(x = AgeGroup, fill = factor(SatisfactionScore))) +
  geom_bar(position = "stack") +
  labs(title = "Satisfaction Scores by Age Group", x = "Age Group", y = "Count", fill =
    "Score") +
  theme_minimal()
ggsave("stacked_bar_satisfaction_by_age.png", width = 7, height = 5)
```

Output



Question 4

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

R Code

```
# 4. Word cloud from open-ended feedback
corpus <- Corpus(VectorSource(df$Feedback))
corpus <- tm_map(corpus, content_transformer(tolower))
corpus <- tm_map(corpus, removePunctuation)
corpus <- tm_map(corpus, removeWords, stopwords("english"))
png("wordcloud_feedback.png", width = 700, height = 500)
wordcloud(corpus, min.freq = 1, colors = brewer.pal(8, "Dark2"), random.order = FALSE)
dev.off()
```

Output

Word Cloud (word-size = frequency) of Customer Feedback



Set 3: Employee Performance Evaluation

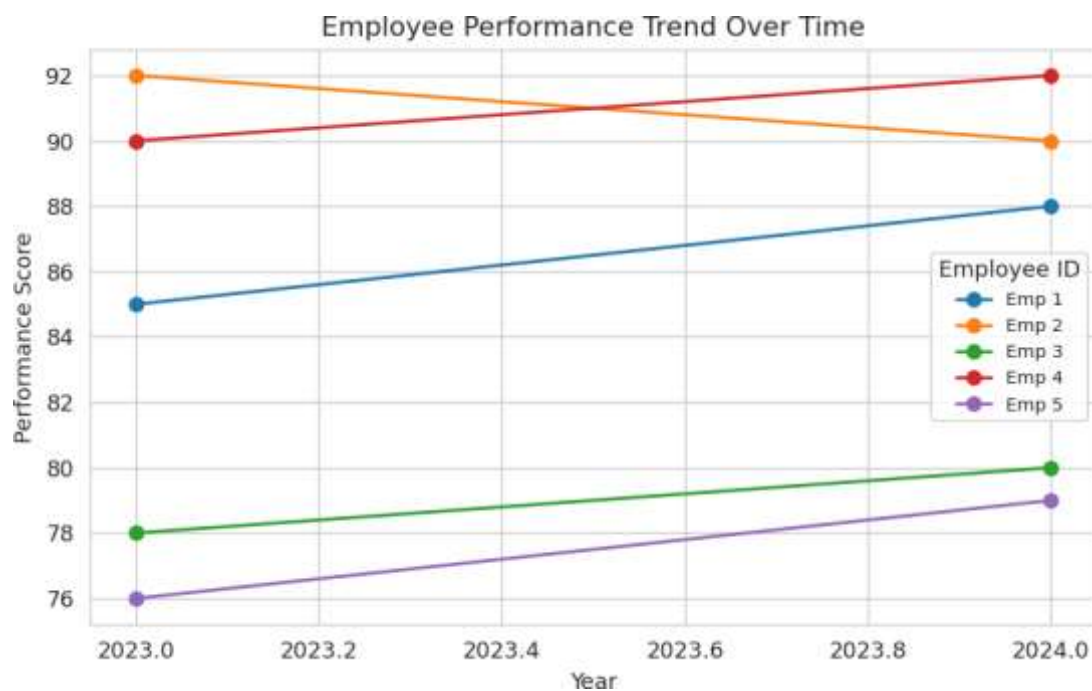
Question 1

Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

R Code

```
# 1. Line chart - performance trend of employees over time
ggplot(df, aes(x = Year, y = PerformanceScore, color = factor(EmployeeID), group = EmployeeID)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(title = "Employee Performance Trend Over Time", x = "Year", y = "Performance Score",
  color = "Employee ID") +
  theme_minimal()
ggsave("line_chart_performance_trend.png", width = 7, height = 5)
```

Output



Question 2

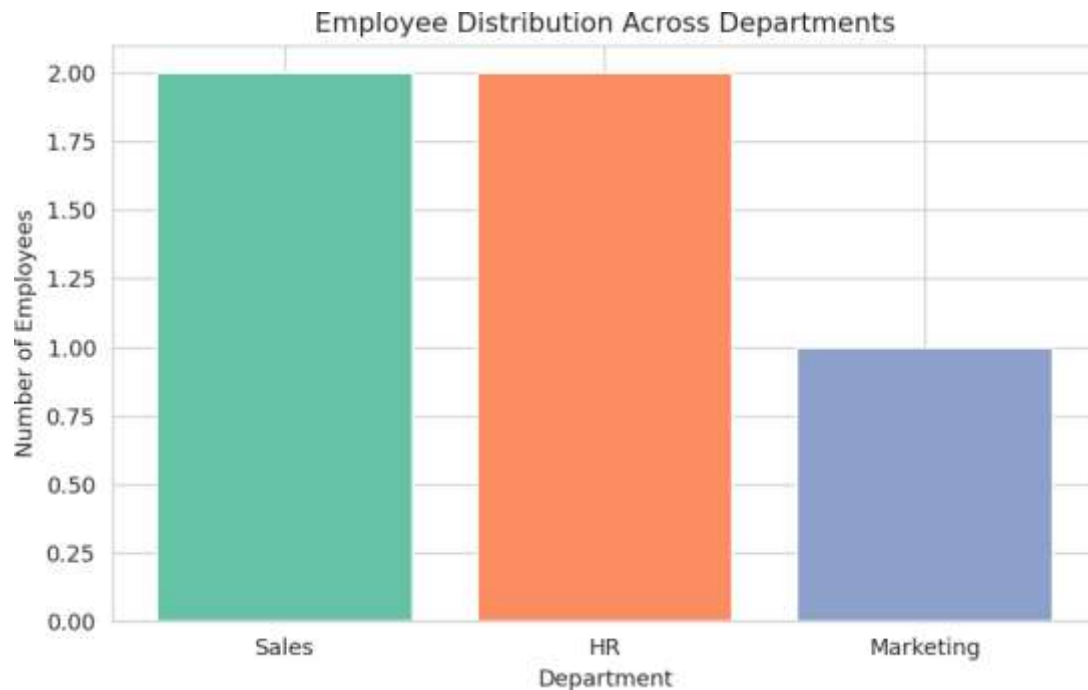
Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

R Code

```
# 2. Bar chart - distribution of employees across departments
dept_counts <- df %>% distinct(EmployeeID, Department) %>% count(Department)
```

```
ggplot(dept_counts, aes(x = Department, y = n, fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Employee Distribution Across Departments", x = "Department", y = "Number
of Employees") +
  theme_minimal()
ggsave("bar_chart_department_distribution.png", width = 7, height = 5)
```

Output



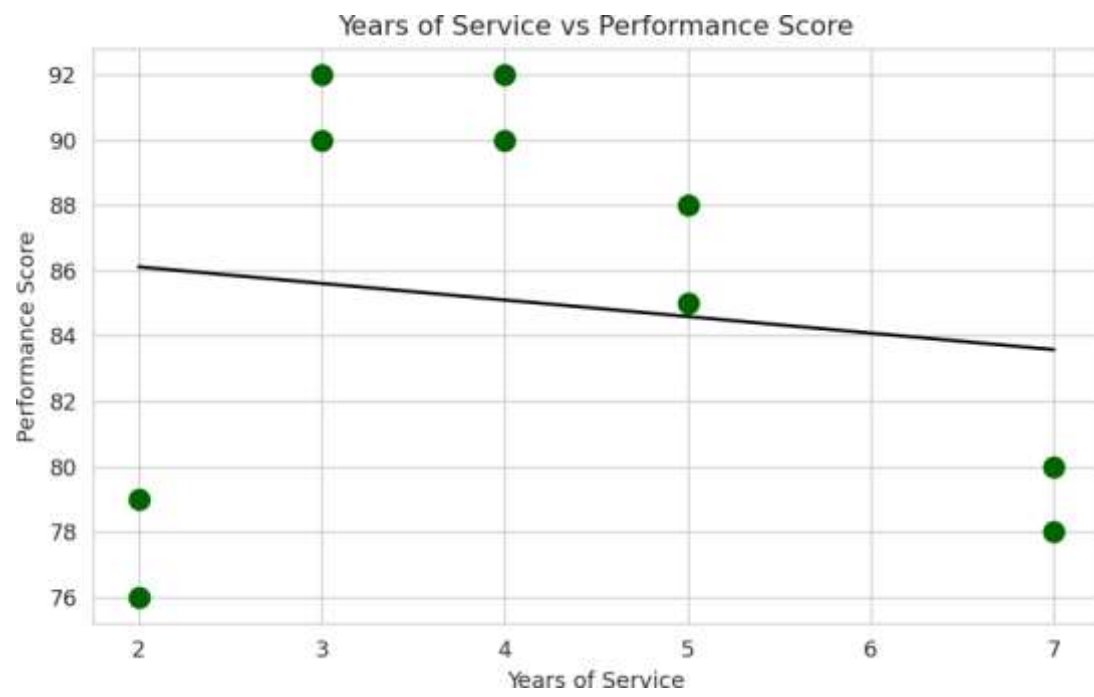
Question 3

Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

R Code

```
# 3. Scatter plot - years of service vs performance score
ggplot(df, aes(x = YearsOfService, y = PerformanceScore)) +
  geom_point(color = "darkgreen", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(title = "Years of Service vs Performance Score", x = "Years of Service", y =
"Performance Score") +
  theme_minimal()
ggsave("scatter_years_vs_performance.png", width = 7, height = 5)
cat("Insight: Correlation coefficient =", cor(df$YearsOfService, df$PerformanceScore),
"\n")
```

Output



Set 4: Product Inventory Management

Question 1

Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

R Code

```
# 1. Bar chart - quantity of each product in inventory
ggplot(df, aes(x = ProductName, y = QuantityAvailable, fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(title = "Quantity of Each Product in Inventory", x = "Product", y = "Quantity
  Available") +
  theme_minimal()
ggsave("bar_chart_inventory_quantity.png", width = 7, height = 5)
```

Output



Question 2

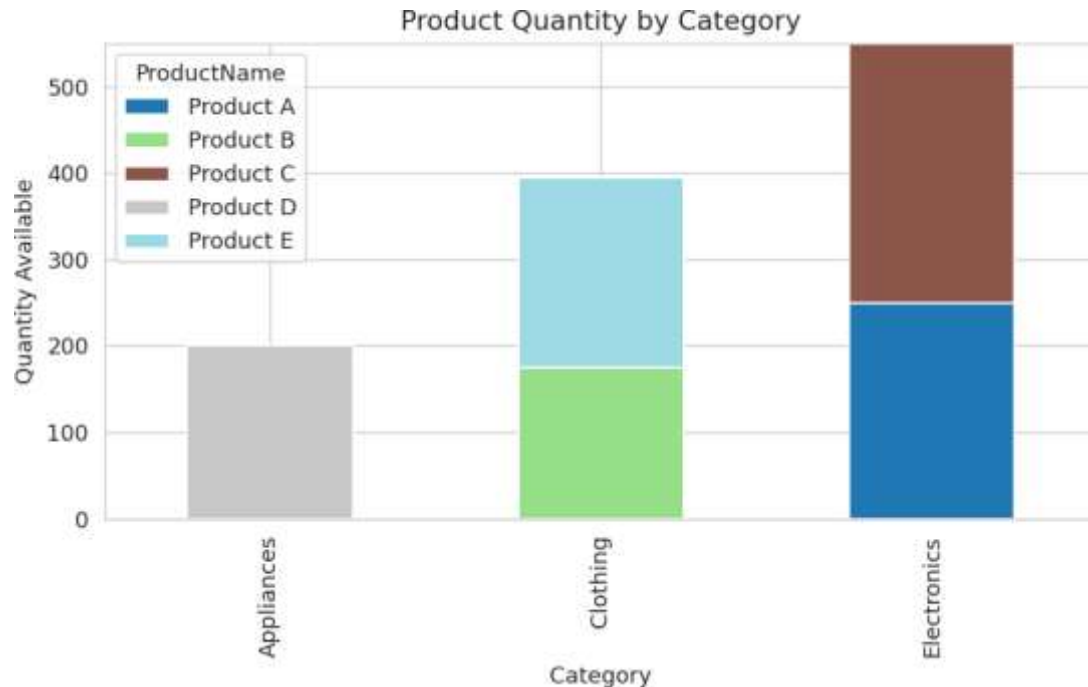
Generate a stacked bar chart to show the quantity of each product within different product categories.

R Code

```
# 2. Stacked bar chart - quantity of each product within categories
ggplot(df, aes(x = Category, y = QuantityAvailable, fill = ProductName)) +
  geom_bar(stat = "identity", position = "stack") +
  labs(title = "Product Quantity by Category", x = "Category", y = "Quantity Available")
+
```

```
theme_minimal()
ggsave("stacked_bar_quantity_by_category.png", width = 7, height = 5)
```

Output



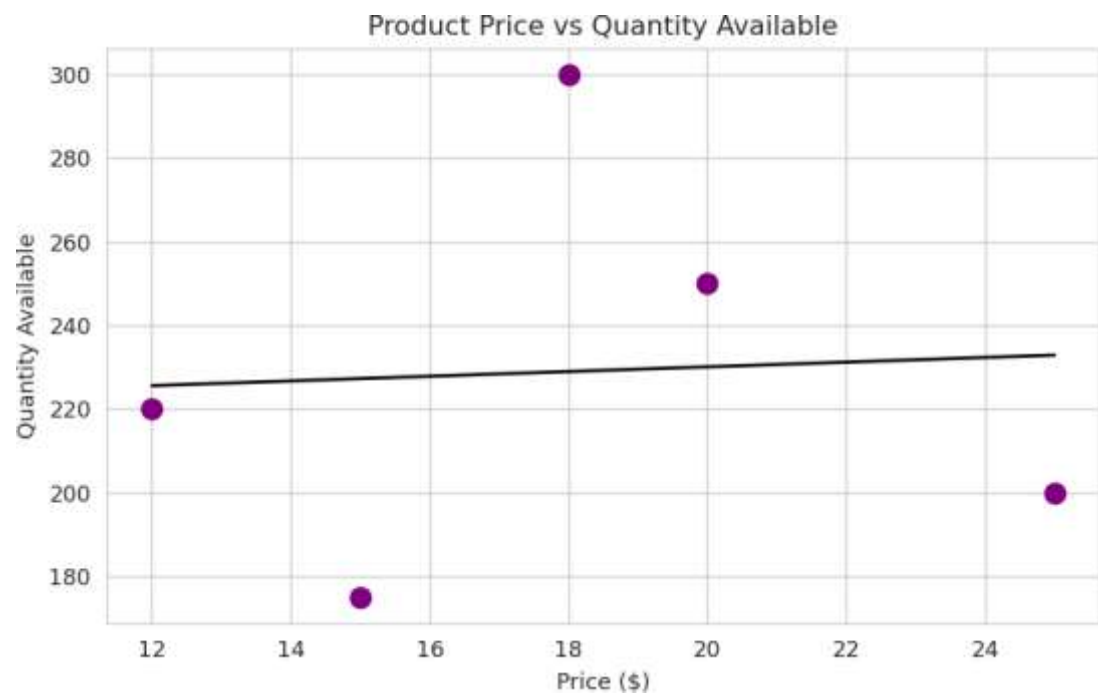
Question 3

Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

R Code

```
# 3. Scatter plot - price vs quantity available
ggplot(df, aes(x = Price, y = QuantityAvailable)) +
  geom_point(color = "purple", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(title = "Product Price vs Quantity Available", x = "Price ($)", y = "Quantity
  Available") +
  theme_minimal()
ggsave("scatter_price_vs_quantity.png", width = 7, height = 5)
cat("Insight: Correlation between price and quantity =", cor(df$Price,
df$QuantityAvailable), "\n")
```

Output



Set 5: Website Analytics

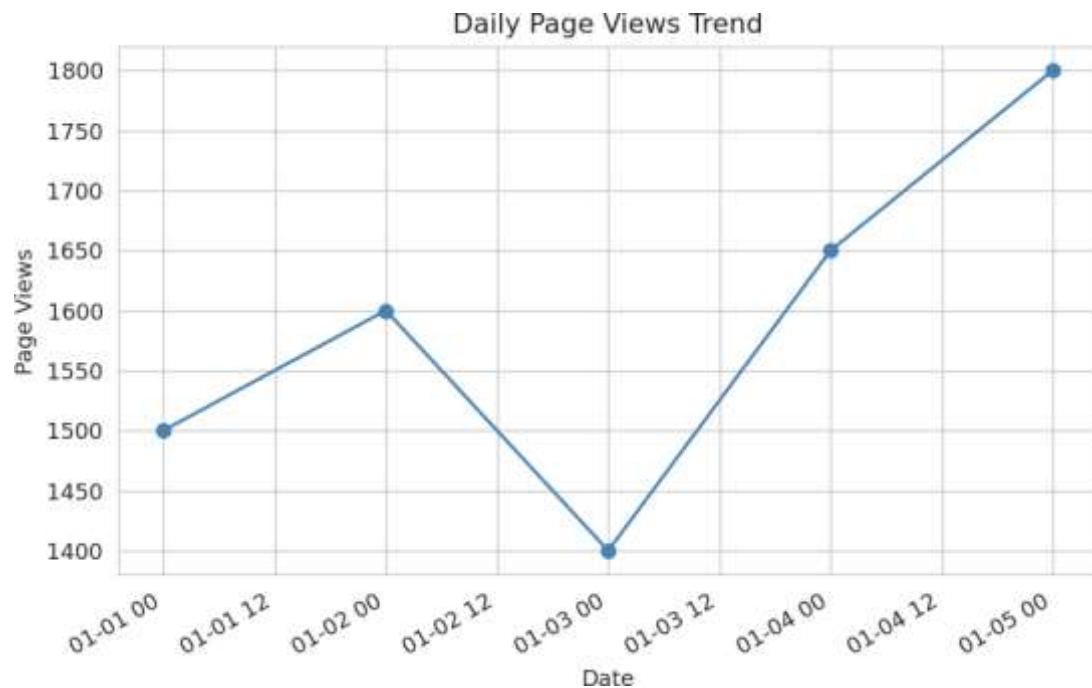
Question 1

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

R Code

```
# 1. Line chart - trend in daily page views
ggplot(df, aes(x = Date, y = PageViews)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_point() +
  labs(title = "Daily Page Views Trend", x = "Date", y = "Page Views") +
  theme_minimal()
ggsave("line_chart_page_views.png", width = 7, height = 5)
```

Output



Question 2

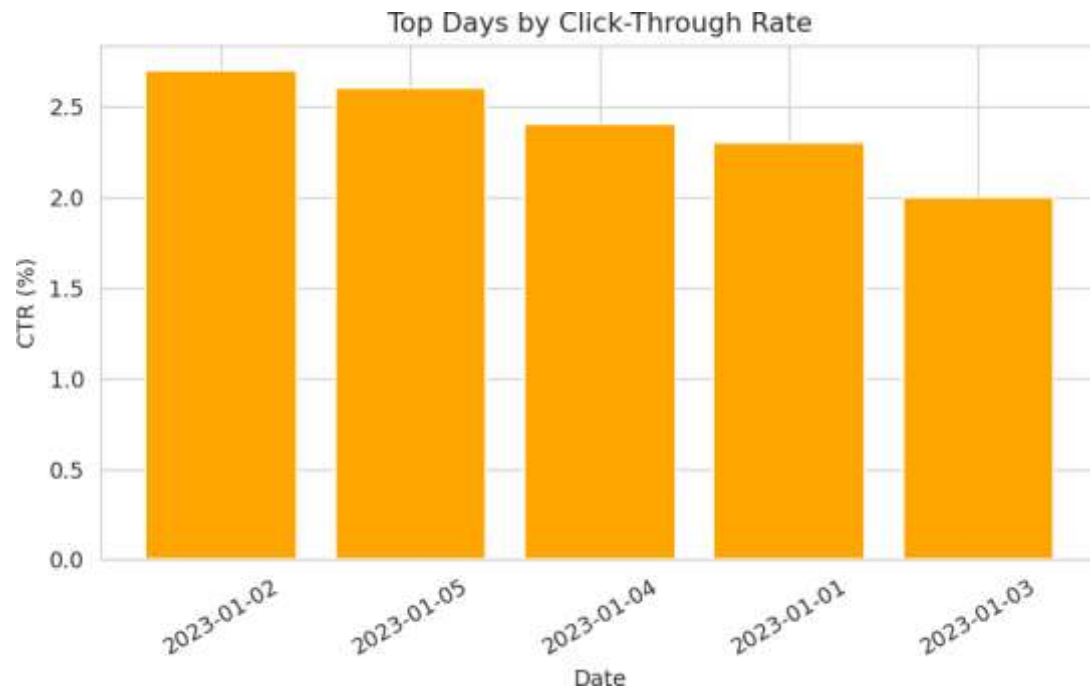
Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

R Code

```
# 2. Bar chart - top N days with highest click-through rates
top_n_days <- df %>% arrange(desc(ClickThroughRate)) %>% head(3)
ggplot(top_n_days, aes(x = reorder(as.character(Date), -ClickThroughRate), y = ClickThroughRate)) +
  geom_bar(stat = "identity", fill = "orange") +
```

```
labs(title = "Top Days by Click-Through Rate", x = "Date", y = "Click-Through Rate (%)") +
theme_minimal()
ggsave("bar_chart_top_ctr_days.png", width = 7, height = 5)
```

Output



Question 3

Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

R Code

```
# 3. Stacked area chart - user interactions (likes, shares, comments)
df_long <- df %>% pivot_longer(cols = c(Likes, Shares, Comments), names_to =
"Interaction", values_to = "Count")
ggplot(df_long, aes(x = Date, y = Count, fill = Interaction)) +
geom_area(position = "stack") +
labs(title = "Website User Interactions Over Time", x = "Date", y = "Count") +
theme_minimal()
ggsave("stacked_area_interactions.png", width = 7, height = 5)
```

Output

